Literature review worksheet

From PICO to search string, search log and evidence matrix

Foundations of Clinical Research · Session 2 (2.1–2.4; proposal Part 1)

Name: *[Your name]*
Research question: *[Copy from proposal 1.4]*
Date: *[DD/MM/YYYY]*

Step 1 From PICO to search terms

Search for P, I (or exposure) and O; C is often left out of the search. For each concept, list synonyms and spelling variants (free-text words, searched in title/abstract [tiab]) AND the controlled vocabulary term (MeSH in PubMed). Combine words WITHIN a concept with OR; combine concepts with AND.

Worked example

	Concept	Synonyms / free text [tiab]	MeSH terms [Mesh]
P	*Septic shock*	*"septic shock", sepsis, "severe sepsis"*	*"Shock, Septic", "Sepsis"*
I	*Restrictive fluids*	*"fluid restriction", restrictive, "fluid balance", "conservative fluid*"*	*"Fluid Therapy", "Water-Electrolyte Balance"*
C	*Usual care*	*(usually not searched)*	*Not searched*
O	*Mortality*	*mortality, death, survival*	*"Mortality", "Hospital Mortality"*

> **Example search string:** *("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab]) AND ("Fluid Therapy"[Mesh] OR "fluid restriction"[tiab] OR restrictive[tiab] OR "fluid balance"[tiab]) AND (mortality[tiab] OR death[tiab] OR "Mortality"[Mesh])*

Our search terms

	Concept	Synonyms / free text [tiab]	MeSH terms [Mesh]
P	*[Concept]*	*[term OR term OR term]*	*[MeSH]*
I / E	*[Concept]*	*[term OR term OR term]*	*[MeSH]*
C	*[Concept]*	*[optional]*	*[MeSH]*
O	*[Concept]*	*[term OR term OR term]*	*[MeSH]*
Design	*[optional filter]*	*[e.g. randomized controlled trial[pt], cohort]*	*[e.g. Cohort Studies]*

Our search string:

[(P terms) AND (I terms) AND (O terms)]

Boolean and search tips

- **AND** narrows (all concepts must appear). **OR** broadens (any synonym). **NOT** excludes. Use it rarely, because it can remove relevant papers.
- "Quotation marks" search an exact phrase; an asterisk truncates (fluid* = fluid, fluids, fluidic). Brackets group the OR terms of each concept.
- Filters (date, species = Humans, article type) are applied after you build the string. Record them in the search log.
- PubMed finds MeSH terms through its MeSH Database; Google Scholar is useful for a quick look but cannot be reproduced. Always do the main search in PubMed (or another bibliographic database).
- Save references in a reference manager (e.g. Zotero, free) from the start.

Step 2 Search log

Record every search so that it can be repeated and reported. "Kept" = papers still relevant after reading titles and abstracts.

#	Database	Date	Search string (or line no.)	Filters	Hits	Kept
1	*PubMed*	*DD/MM/YYYY*	*Example string above*	*Humans; last 10 years*	*n*	*n*
2	*[Database]*	*[Date]*	*[String]*	*[Filters]*		
3	*[Database]*	*[Date]*	*[String]*	*[Filters]*		
4	*[Database]*	*[Date]*	*[String]*	*[Filters]*		
5	*[Database]*	*[Date]*	*[String]*	*[Filters]*		

Step 3 Evidence matrix

One row per key paper. Main result = the effect size with its 95% confidence interval (e.g. RR, OR, HR, mean difference), not only the p-value. The first two rows are worked examples from the course papers.

#	Author, year	Design	Population and setting	n	Main result (effect size, 95% CI)	Limitations	Relevance to our question
1	*Boulet et al., 2024 (Crit Care)*	*Pilot RCT, single-blind, 3 ICUs*	*Adults 18–85 y with septic shock, first 24 h of vasopressors*	*48 analysed (24 / 24)*	*Cumulative fluid balance at day 5: median 6.9 vs 35.0 ml/kg; difference 13.2 ml/kg (95% CI −15.2 to 41.6), p = 0.42. 28-day mortality similar.*	*Small pilot; clinicians not blinded; primary outcome is fluid balance, not mortality*	*Same PICO: restriction is possible, but the effect on mortality is unknown*
2	*Hyun et al., 2023 (Ann Intensive Care)*	*Prospective registry cohort, propensity-score matched, 20 hospitals*	*Adults with sepsis admitted to ICU*	*2516 eligible; 373 matched pairs (day-2 cohort)*	*Positive vs negative cumulative balance to day 2: 28-day mortality HR 2.13 (95% CI 1.48–3.06). Day 1: HR 1.17 (0.85–1.60).*	*Observational: residual confounding by indication; selection into cohorts*	*Positive balance is associated with death; cannot show that restriction helps*
3							
4							
5							
6							
7							

Step 4 What is known, and the gap

Summarise the matrix in three short paragraphs. This becomes the problem statement and rationale (proposal 1.2).

What is known

[Write here]

What is NOT known (the gap)

[Write here]

How our study will help fill the gap

[Write here]

Step 5 Five key references (Vancouver style)

Authors (up to 6, then et al.). Title. Journal abbreviation. Year;Volume(Issue):Pages. doi

> **Example:** *Hyun DG, Ahn JH, Huh JW, et al. Impact of a cumulative positive fluid balance during the first three ICU days in patients with sepsis: a propensity score-matched cohort study. Ann Intensive Care. 2023;13:105. doi:10.1186/s13613-023-01178-x*

1. *[Reference 1]*
2. *[Reference 2]*
3. *[Reference 3]*
4. *[Reference 4]*
5. *[Reference 5]*